

Extremum problems

Lessons 23–25 · Algebra 11, §1.8 · P1 · 7.8 · 28 problems

Name the variable, write the quantity to be optimised in terms of that one variable, then differentiate. Use the constraint to remove the second variable **before** differentiating, and state the units in the answer.

Easy 7

The modelling is done for you, or takes one line. Problem 6 is about the domain, which is part of the answer and not an afterthought.

- 1 Two numbers add to 30. Maximise their product. 225

$P = x(30 - x) = 30x - x^2$, so $P' = 30 - 2x = 0$ gives $x = 15$ and the other number is also 15.

- 2 Two numbers add to 10. Minimise the sum of their squares. 50

$S = x^2 + (10 - x)^2$, so $S' = 2x - 2(10 - x) = 4x - 20 = 0$ gives $x = 5$.

- 3 Maximum value of $y = 12x - x^2$ 36, at $x = 6$

$y' = 12 - 2x = 0$.

- 4 Minimum value of $y = x^2 - 8x + 3$ -13, at $x = 4$

$y' = 2x - 8 = 0$, then $y(4) = 16 - 32 + 3$.

- 5 200 m of fence makes a rectangular field. Find the greatest area. 2500 m²

Two sides x and two sides $100 - x$, so $A = x(100 - x)$ and $x = 50$. The best rectangle is a square.

- 6 A square of side x is cut from each corner of a 16×16 card. Give the domain of x . $0 < x < 8$

The cut cannot be zero and two cuts must fit along a side, so $2x < 16$.

- 7 A rectangle has perimeter 24. Write its area in terms of one side x . $A = x(12 - x)$

The other side is $\frac{24 - 2x}{2} = 12 - x$.

Medium 7

One constraint to substitute. In 12 the greatest value sits at an endpoint, not at a stationary point, which is why the ends must be tested.

- 8 60 m of fence encloses a rectangle against a wall (three sides). Find the greatest area. 450 m²

With x the side parallel to the wall, $x + 2y = 60$, so $A = x \cdot \frac{60 - x}{2}$. Then $A' = 30 - x = 0$ gives $x = 30$ and $y = 15$.

- 9 Corner squares of side x are cut from a 24×24 card and the sides folded up. Find the greatest volume.

1024 cm³, at $x = 4$

$V = x(24 - 2x)^2$. Differentiating, $V' = (24 - 2x)(24 - 6x)$, so $x = 4$ or $x = 12$. Only $x = 4$ lies in the domain $0 < x < 12$.

- 10 A closed cylinder holds 2000 cm^3 . Find the radius that minimises the surface area.

$$r = \sqrt[3]{\frac{1000}{\pi}} \approx 6.83 \text{ cm}$$

$$h = \frac{2000}{\pi r^2}, \text{ so } S = 2\pi r^2 + 2\pi rh = 2\pi r^2 + \frac{4000}{r}. \text{ Then } S' = 4\pi r - \frac{4000}{r^2} = 0 \text{ gives } r^3 = \frac{1000}{\pi}, \text{ and } S \approx 879 \text{ cm}^2.$$

- 11 Least value of $y = x + \frac{16}{x}$ for $x > 0$ **8, at $x = 4$**

$$y' = 1 - \frac{16}{x^2} = 0 \text{ gives } x^2 = 16, \text{ and only } x = 4 \text{ is in range.}$$

- 12 Greatest value of $y = x^3 - 6x^2$ on $[0, 5]$ **0, at $x = 0$**

$$y' = 3x(x - 4) \text{ gives } x = 0 \text{ and } x = 4. \text{ The candidates are } y(0) = 0, y(4) = -32, y(5) = -25. \text{ The greatest is at an endpoint.}$$

- 13 A rectangle has area 64 cm^2 . Minimise its perimeter. **32 cm**

$$P = 2(x + \frac{64}{x}), \text{ so } P' = 2(1 - \frac{64}{x^2}) = 0 \text{ gives } x = 8 \text{ — a square again.}$$

- 14 An open box has a square base of side x and volume 500 cm^3 . Write its surface area in terms of x .

$$S = x^2 + \frac{2000}{x}$$

$$h = \frac{500}{x^2}, \text{ and the box has one base and four sides: } S = x^2 + 4xh.$$

Hard 7

The model takes several lines. In 15 and 17 a geometric relation is the constraint; in 16 minimising the square of the distance avoids the root and gives the same answer.

- 15 A rectangle is inscribed in a circle of radius 10. Maximise its perimeter. **$40\sqrt{2} \approx 56.6$**

$$\text{With sides } x \text{ and } y, x^2 + y^2 = 400, \text{ so } y = \sqrt{400 - x^2} \text{ and } P = 2x + 2\sqrt{400 - x^2}. \text{ Then } P' = 2 - \frac{2x}{\sqrt{400 - x^2}} = 0 \text{ gives } x = \sqrt{400 - x^2}, \text{ so } x = y = 10\sqrt{2} \text{ — a square.}$$

- 16 Find the point on $y = 2x + 1$ closest to the origin. **$(-\frac{2}{5}, \frac{1}{5})$, distance $\frac{\sqrt{5}}{5}$**

$$\text{Minimise } D = x^2 + (2x + 1)^2 \text{ rather than the distance itself; the same } x \text{ minimises both, because the square root is increasing. } D' = 10x + 4 = 0 \text{ gives } x = -\frac{2}{5}.$$

- 17 A cylinder is inscribed in a sphere of radius 3. Maximise its volume. **$12\sqrt{3}\pi \approx 65.3$**

$$\text{With height } h, \text{ the radius satisfies } r^2 + (\frac{h}{2})^2 = 9, \text{ so } V = \pi r^2 h = \pi(9 - \frac{h^2}{4})h. \text{ Then } V' = 9\pi - \frac{3\pi h^2}{4} = 0 \text{ gives } h^2 = 12, \text{ so } h = 2\sqrt{3} \text{ and } r^2 = 6.$$

- 18 A wire 100 cm long is cut into two pieces, one bent into a square and the other into an equilateral triangle. Minimise the total area.

$\approx 272 \text{ cm}^2$, with $\approx 43.5 \text{ cm}$ in the square

With x cm in the square, its area is $\frac{x^2}{16}$; the triangle has side $\frac{100-x}{3}$ and area $\frac{\sqrt{3}}{4}(\frac{100-x}{3})^2$. Differentiating and setting to zero gives $x = \frac{800\sqrt{3}}{18+8\sqrt{3}} \approx 43.50$. Cutting nothing off — all wire in the square — gives 625 cm^2 , so the stationary point really is the least.

- 19 An open-topped box with a square base holds 4000 cm^3 . Minimise the material used.

1200 cm^2 , with base 20 cm and height 10 cm

$S = x^2 + 4xh$ with $h = \frac{4000}{x^2}$, so $S = x^2 + \frac{16000}{x}$. Then $S' = 2x - \frac{16000}{x^2} = 0$ gives $x^3 = 8000$.

- 20 A running track is a rectangle with a semicircular end at each side, of total perimeter 400 m. Maximise the area of the rectangle.

$\frac{20000}{\pi} \approx 6366 \text{ m}^2$

With r the radius of the semicircles and x the length of the rectangle, the perimeter is $2x + 2\pi r = 400$, so $x = 200 - \pi r$. The rectangle is x by $2r$, so $A = 2r(200 - \pi r)$. Then $A' = 400 - 4\pi r = 0$ gives $r = \frac{100}{\pi}$ and $x = 100 \text{ m}$.

- 21 A box with a square base and no lid is made from 300 cm^2 of card. Maximise its volume.

500 cm^3 , base 10 cm, height 5 cm

$x^2 + 4xh = 300$, so $h = \frac{300 - x^2}{4x}$ and $V = x^2h = 75x - \frac{x^3}{4}$. Then $V' = 75 - \frac{3x^2}{4} = 0$ gives $x^2 = 100$.

Very hard 7

Letters instead of numbers, or a constraint that has to be built from a diagram. In 24 the numbers are chosen so that the root comes out exactly.

- 22 Show that among all rectangles of a given perimeter, the square has the greatest area. shown

Let the perimeter be p , so the sides are x and $\frac{p}{2} - x$. $A = x(\frac{p}{2} - x)$, and $A' = \frac{p}{2} - 2x = 0$ gives $x = \frac{p}{4}$ — so both sides are $\frac{p}{4}$. $A'' = -2 < 0$ confirms a maximum, and the greatest area is $\frac{p^2}{16}$.

- 23 A cylinder is inscribed in a cone of height 12 and base radius 5, with its base on the base of the cone.

Maximise its volume. $\frac{400\pi}{9} \approx 140$, with $r = \frac{10}{3}$ and $h = 4$

Similar triangles give $\frac{h}{12} = \frac{5-r}{5}$, so $h = 12 - \frac{12r}{5}$. Then $V = \pi r^2h = \pi(12r^2 - \frac{12r^3}{5})$ and $V' = \pi(24r - \frac{36r^2}{5}) = 0$ gives $r = \frac{10}{3}$.

- 24 A pipeline runs from a rig 5 km offshore to a refinery 13 km along a straight coast. Underwater pipe costs \$5m per km and land pipe \$3m per km. Where should the pipe meet the shore, and what is the least cost? **3.75 km from the nearest point; \$59m**

With x km along the shore from the point nearest the rig, $C = 5\sqrt{25 + x^2} + 3(13 - x)$. Then $C' = \frac{5x}{\sqrt{25 + x^2}} - 3 = 0$ gives $5x = 3\sqrt{25 + x^2}$, so $25x^2 = 225 + 9x^2$ and $x = \frac{15}{4}$. Then $\sqrt{25 + x^2} = 6.25$ and $C = 31.25 + 27.75 = 59$. Going straight ashore costs $25 + 39 = 64$, and going straight to the refinery costs $5\sqrt{194} \approx 69.6$, so the compromise really is cheaper.

- 25 The height plus the circumference of a cylindrical parcel may not exceed 300 cm. Maximise its volume.

$$\frac{1000000}{\pi} \approx 318000 \text{ cm}^3, \text{ with } r = \frac{100}{\pi} \text{ and } h = 100 \text{ cm}$$

The largest parcel uses the whole allowance, so $h + 2\pi r = 300$ and $V = \pi r^2(300 - 2\pi r)$. Then $V' = 600\pi r - 6\pi^2 r^2 = 6\pi r(100 - \pi r) = 0$ gives $r = \frac{100}{\pi} \approx 31.8$ cm, and $h = 300 - 200 = 100$ cm.

- 26 Show that the rectangle of greatest area inscribed in a semicircle of radius r , with one side on the diameter, has area r^2 . **shown**

Corners at $(\pm x, y)$ with $x^2 + y^2 = r^2$, so $A = 2x\sqrt{r^2 - x^2}$. Maximise $A^2 = 4x^2(r^2 - x^2)$ instead: differentiating gives $8xr^2 - 16x^3 = 0$, so $x^2 = \frac{r^2}{2}$ and then $y^2 = \frac{r^2}{2}$. Hence $A = 2 \cdot \frac{r}{\sqrt{2}} \cdot \frac{r}{\sqrt{2}} = r^2$. The rectangle is twice as wide as it is tall, so it is not a square.

- 27 Find the point on $y = x^2$ closest to $(3, 0)$. **(1, 1), distance $\sqrt{5}$**

Minimise $D = (x - 3)^2 + x^4$. Then $D' = 2(x - 3) + 4x^3 = 0$, that is $2x^3 + x - 3 = 0$, which factorises as $(x - 1)(2x^2 + 2x + 3) = 0$. The quadratic has discriminant $4 - 24 < 0$, so $x = 1$ is the only real root — one closest point, not two.

- 28 A cylindrical can with no lid has volume V . Show that the material is least when the height equals the radius. **shown**

$S = \pi r^2 + 2\pi rh$ with $h = \frac{V}{\pi r^2}$, so $S = \pi r^2 + \frac{2V}{r}$. Then $S' = 2\pi r - \frac{2V}{r^2} = 0$ gives $r^3 = \frac{V}{\pi}$, that is $V = \pi r^3$. Putting that back, $h = \frac{\pi r^3}{\pi r^2} = r$. $S'' = 2\pi + \frac{4V}{r^3} > 0$, so it is a minimum. A closed can would give $h = 2r$ instead — the lid changes the answer.